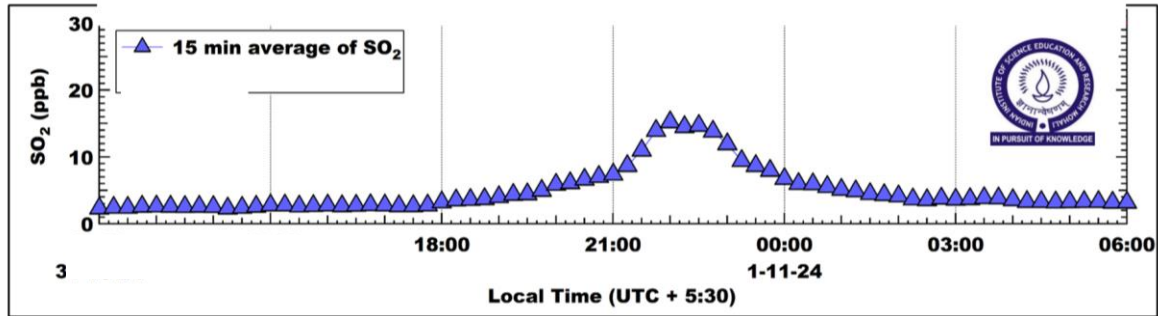


Q1) Calculate the number density of 400 ppm carbon dioxide at sea level for  $P = 1013 \text{ hPa}$ ,  $T = 0 \text{ }^\circ\text{C}$ . Also calculate its mass concentration.

Q2) On Diwali night 2024 at IISER Mohali the following values for observed for time series of sulphur dioxide. If the national ambient air quality standard for  $\text{SO}_2$  is  $80 \mu\text{g m}^{-3}$ , determine whether and for how long the ambient values crossed it.



Q3) An air quality station recorded the following daily average values in ambient air: air temperature =  $300 \text{ K}$ ; air pressure =  $960 \text{ mbar}$ ;  $\text{NO}_2 = 90 \text{ ppbV}$ . If the daily ambient air quality standard for  $\text{NO}_2$  is  $80 \mu\text{g/m}^3$ , determine whether the site complies with the ambient air quality standard for  $\text{NO}_2$ . ( $1 \text{ mbar} = 100 \text{ Pa}$ ; M.W. of  $\text{NO}_2 = 46 \text{ g/mol}$ )

Q4) A hypothetical planet "MARGA999" containing an atmosphere like Earth has a diameter of  $1000 \text{ km}$  and "g" value of  $20 \text{ m/s}^2$ . If the surface pressure at Surya999 is  $500 \text{ hPa}$ , calculate the mass of the atmosphere on Surya999.

Q5) If a massive fire in the global peatlands increases the global average atmospheric CO by  $1000 \text{ ppb}$  on planet Earth over a period of a full year, calculate in  $\text{TgC}$ , the amount of carbon emitted per day into the atmosphere due to this fire alone in that year? Clearly state the assumptions made in your calculations (Radius of Earth =  $6400 \text{ km}$ ; Mean surface atmospheric pressure =  $984 \text{ hPa}$ ;  $g = 9.8 \text{ m s}^{-2}$ )

Q6) An air quality station recorded the following 8h daytime average value in ambient air: air temperature =  $300 \text{ K}$ ; air pressure =  $960 \text{ mbar}$ ;  $\text{O}_3 = 52 \text{ ppb}$ . If the ambient 8h daytime average air quality standard for  $\text{O}_3$  is  $100 \mu\text{g/m}^3$ , determine whether the site complies with the ambient air quality standard for  $\text{O}_3$ . ( $1 \text{ mbar} = 100 \text{ Pa}$ ; M.W. of  $\text{O}_3 = 48 \text{ g/mol}$ )

Q7) ) An air quality station recorded the following 8h daytime average value in ambient air: air temperature =  $300 \text{ K}$ ; air pressure =  $1000 \text{ mbar}$ ;  $\text{CO} = 1000 \text{ ppb}$ . If the ambient 8 h average air quality standard for CO is  $2 \text{ mg/m}^3$ , determine whether the site complies with the ambient air quality standard for CO. ( $1 \text{ mbar} = 100 \text{ Pa}$ ; M.W. of CO =  $28 \text{ g/mol}$ )

Q8) Show that an increase of  $1.8 \text{ ppmV CO}_2/\text{yr}$  in the Earth's atmosphere corresponds to an increase of  $4 \times 10^{15} \text{ gC}$  per year.

Q9) Convert  $10 \text{ ppb}$  of benzene at a temperature of  $300 \text{ K}$  and  $1 \text{ atmosphere}$  pressure to micrograms of benzene per  $\text{m}^3$  of air.

Q10) Calculate the settling velocity of  $\text{PM}_{2.5}$  and also estimate its atmospheric residence time against deposition for  $\text{PM}_2$  released from  $100 \text{ m}$  stack height.